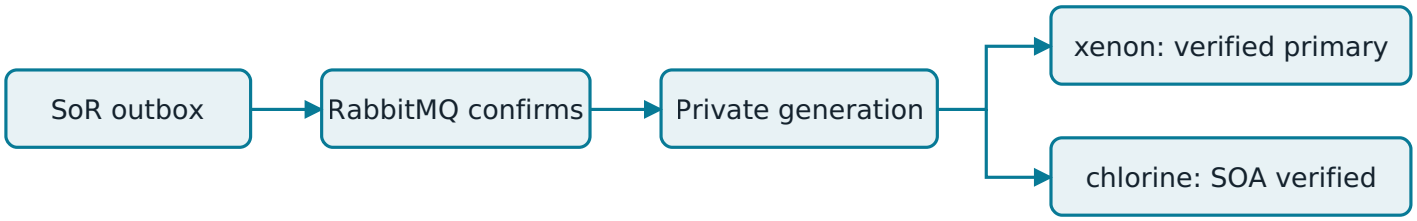
DNS delivery and recovery

SolariNet · DNS operations manual · 21 October 2026

Candidate boundary: The new multi-target SoR release applier remains uninstalled; the legacy live SoR applier is still active. The candidate now verifies Xenon first, then refreshes each named secondary and requires SOA parity before broker acknowledgment. It has no direct copy-to-Steel path; the bare-zone generator defaults to the active chlorine target only. Separately, BIND 9.18.39 on the Khadas VIM4 `chlorine` (`10.7.0.10`) runs as an authoritative-only secondary for 109 zones: `akoria.net`, 17 generated reverse zones, 90 single-label zones, and `rpz.akoria`. A primary refresh reports current SOA serials for all 109 zones and Radium's forward zone; canonical BIND hashes match for all 109 zones. Recursion is refused, and the `step-ca` service remains active. Xenon watches all primary zone fragments and the RPZ data file, with a five-minute retry timer. The user's Steel label is on a ZimaBlade at port 12 of `laboratory`; the user reports it booted without useful output. Current network checks show no route to it, so I cannot image it remotely. The user also has a ZimaBoard and a ZimaBoard 2. The ZimaBlade's tertiary DNS role awaits console redeployment; it is not configured as a nameserver or listed in the NS set.

1. Delivery contract



The emitter advances its checkpoint only after mandatory routing and broker confirmation. The DNS consumer renders current SoR state into a private directory and first verifies the Xenon release. It then refreshes the generated secondary-zone list on every configured secondary and waits for SOA parity across all generated zones. It acknowledges only after each target proves serving and its local status record is durable. An unchanged primary does not suppress a secondary retry.

Target work runs in a worker while the RabbitMQ owner thread handles heartbeats. After connection loss, the in-flight operation finishes before a retry starts. Startup and hourly reconciliation recover drift or missed events. The emitter checkpoint alone does not prove target freshness.

2. Reviewed installation and cutover

1. **Prepare the primary and configured secondaries.** Back up BIND config, zone data, release history, and journals. Verify restore. Stop the old copy-based DNS writer and confirm it is inactive; its lock does not participate in the new protocol.
2. **Install protected artifacts.** Install `deploy/dns-release/release.py` as root-owned code at `/usr/local/libexec/solarinet-dns-release`. Install the secondary zone updater on each configured secondary. Review the matching JSON example and install it at `/etc/solarinet/dns-release.json`, owned by root and not writable by other accounts. Confirm Xenon host/container path mappings and secondary transfer ACLs.
3. **Prepare the applier account.** Create private mode-0700 incoming and state directories owned by the configured `candidateUser`. The examples use the observed legacy account `jason`; the F-04 dedicated-account cutover remains required. Give that account only the reviewed helper capability.

4. **Bootstrap each managed include.** The helper snapshots only configured primary zones, validates full config and current serving, then atomically replaces their parent include. Existing legacy zone files stay intact. The parent backup and cutover journal are private.

```
sudo /usr/local/libexec/solarinet-dns-release --bootstrap \
  --config /etc/solarinet/dns-release.json
```

Successful conversion replaces the configured legacy includes with one include of `solarinet-current/named.conf.managed`. Unmanaged RPZ and AD-forward zones remain outside the generation. A changed legacy source during preparation aborts cutover.

- **5. Update the SoR applier.** Use the new `[dns]` keys in `deploy/sorsync/sorsync.conf.example`: private `incoming_root`, `target_state_file`, applied-zone directory, primary release helper/config, and secondary refresh helper/config. Add one `[dns.secondary.<name>]` section per provisioned secondary. Direct Steel file-copy settings are removed; Steel is not configured while its ZimaBlade is offline. Install `deploy/sorsync/requirements.txt` into the service venv and set `render_python` to that same `.venv/bin/python3`; PyMySQL and PyYAML are both required by the SoR rendering path.
- **6. Start and verify.** Start the new applier only after the primary bootstrap and every configured secondary refresh succeed. Confirm each target's authoritative SOA and expected records, local target state, queue convergence, and recovery after a controlled target outage. Verify actual secondary AXFR and rollback before release.

3. Read status precisely

Field / file	Meaning
`targets.json` → `desiredSource`	Hash of the rendered input requested for that target.
`appliedSource`, `appliedGeneration`	The input last proved applied, and the root helper's actual immutable serving artifact. Target serial rebasing can make these hashes differ.
`phase`, `lastAttempt`, `lastSuccess`	Pending/applied/failed/disabled state and timestamps. A failed target retains its prior successful generation and time.
Root `state.json`	Desired/applied generation, activation phase, serial high-water marks, and last verified success.
Root `bootstrap.json`	Initial include conversion and recovery journal. A pending conversion blocks later data activation.

Primary success with a secondary failure is partial convergence: the event remains unacknowledged, that secondary stays failed/pending, and every configured secondary is attempted again. The ZimaBlade's future tertiary role is enabled only by adding a named secondary section after redeployment and independent service verification.

4. Recovery and retention

Failed activation: the helper revalidates the prior records, creates a rollback generation with serials newer than the attempted release, switches, reloads, and verifies serving. This allows secondaries that already transferred the failed generation to follow the rollback. Manually pointing at an older serial is insufficient.

Interrupted activation or failed rollback: the journal retains the previous and rollback identities. The next helper invocation retries recovery before new activation. The rollback generation is reused during retries. Removed/reintroduced zones retain serial history.

Interrupted bootstrap: re-run `--bootstrap`. It restores the exact legacy parent contents, mode, and ownership before attempting conversion again. Do not delete journals to bypass recovery. Keep previous generations, backups, and serial high-water marks until a reviewed retention policy covers all secondaries and replay sources.

5. Evidence and remaining gates

Exercised: five isolated renderer tests; 17 real BIND activation/cutover tests; and 37 SoR-sync tests against isolated MariaDB 11 and RabbitMQ 4. The chain test inserts CDC-tracked entities and IP addresses, runs the real publisher and SoR DNS renderer, invokes the release CLI through `sudo -n` with a root-owned temporary config and release tree, and checks the served primary answer plus persisted primary and secondary target state before each broker acknowledgment. A private loopback BIND process running as the actual `bind` service account serves the primary zones; the secondary refresh is a controlled helper stub, not live chlorine. The release suite includes a restrictive-umask activation check; generated zone directories remain readable by BIND. Zone verification accepts NS-only zones and prefers A/PTR checks when available. Broker cases cover slow apply beyond heartbeat intervals, redelivery after failed apply, mandatory-route failure before checkpoint advancement, recovery after restarting the RabbitMQ application, and recovery after stopping and restarting the disposable broker container. The DNS consumer re-registers, reconciles, consumes a confirmed event, and drains its fixture queue after either restart.

Observed live on 2026-10-21 at 07:35 UTC: Xenon's `named-checkconf -z` loaded the forward, reverse, bare-name, and RPZ zones. The legacy SoR applier remains active; the candidate release helper/config are absent. The Khadas service reports current SOA serials for 109 zones plus Radium's forward zone. Canonical `named-compilezone` hashes match on all 109 zones, including SOAs. Direct queries returned authoritative SOAs for `dashboard` (`2026071001`) and `rpz.akoria` (`2026102101`); the `akoria.net` NS set includes chlorine, and recursion returns `REFUSED`. BIND listens only on `10.7.0.10:53`; `step-ca` remains active. Xenon's ACLs allow chlorine transfers for bare zones and RPZ; their zone data was not rewritten. At this earlier observation, the legacy generator was configured with both chlorine and the reserved ZimaBlade transfer addresses. The release candidate now defaults to chlorine alone and permits explicit configured secondaries; its isolated generator tests cover serial preservation and updates. The SoR-to-DNS scratch chain tests CDC, confirmed publication, rendering, release-engine activation, and authoritative answers on a private BIND process before acknowledgment; its secondary refresh is an isolated helper stub, not live chlorine. The ZimaBlade at `10.0.0.11` is unreachable from Xenon (ping loss; SSH `No route to host`); tertiary DNS awaits console redeployment. Radium serves the forward zone only. Cesium hosts SoR MariaDB; RabbitMQ runs on benzene.

Unverified: candidate SoR release cutover, account separation, crash/restore drills, client resolver failover, and cross-lab review. Production SoR delivery has not yet used the candidate release helper. BIND views are refused; rendering targets IPv4. Pi-hole/AD integration needs live drills. Zima remains offline and is not configured as a nameserver.

Single-file HTML/CSS/browser-JS/inline-SVG manual. The parallel Letter PDF is generated from this source. Live legacy DNS/secondary state is reported separately from the uninstalled candidate SoR release applier.